

Name: Ahmed Wael Adel

ID: 22P0309

Date / /

Subject

11.75

20

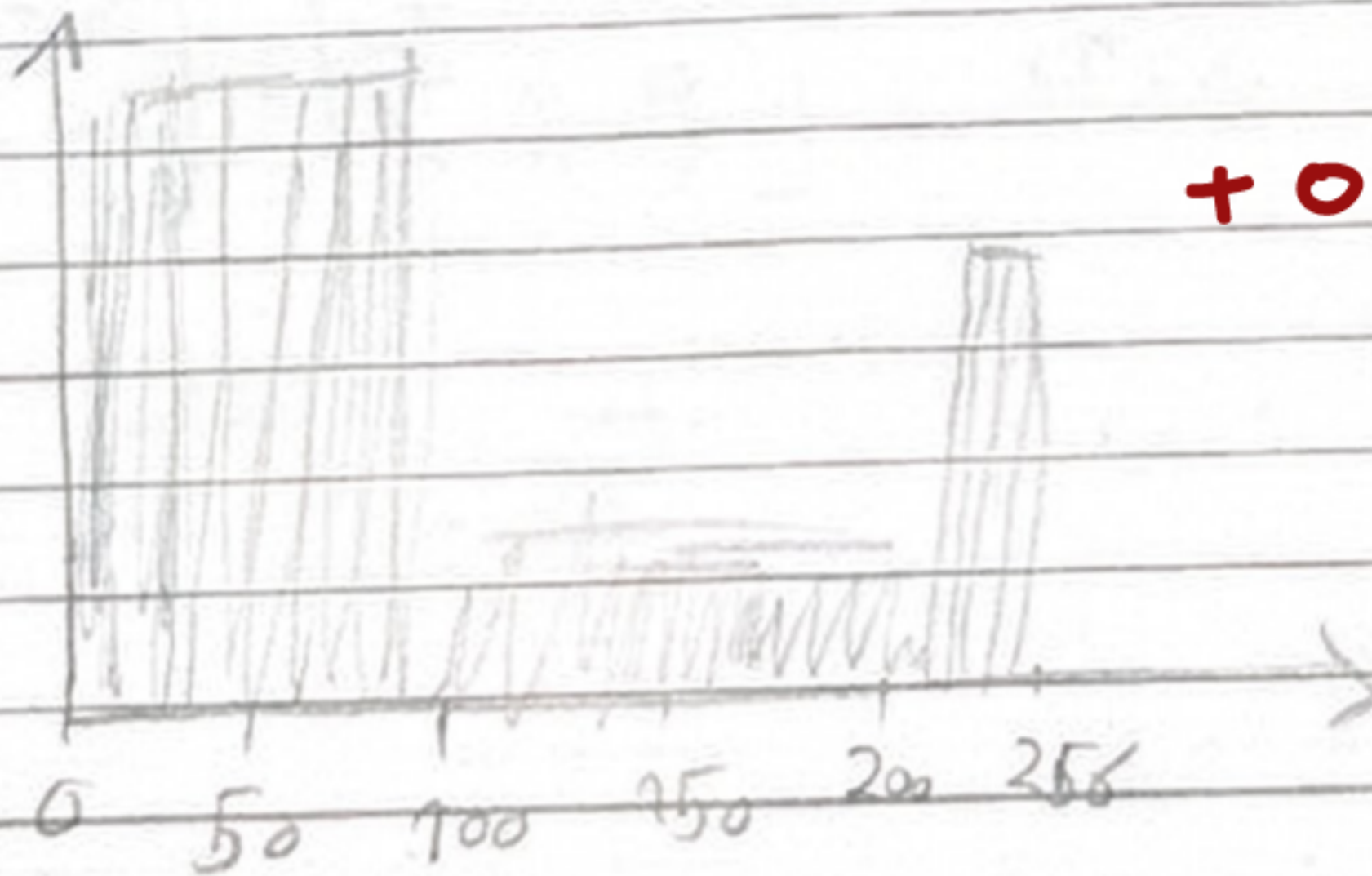
Q(1)

* Intensity Values = 20, 25, 50, 15, 25
+ 0.5

* Comment on classification

$\frac{1}{4}$

* Binarization?
Intensity th ? - 3
Methods



+ 0.5

Histogram

Q(2)

$$\therefore L_1 = 4, L_2 = 6$$

$$\frac{L_2}{L_1} = \frac{6}{4} = 1.5$$

$$\text{Scaling} \rightarrow \times 1.5 \quad A(3, 3) \xrightarrow{\times 1.5} A'(4.5, 4.5)$$

But the in Real, $A'(6.5, 3.4)$

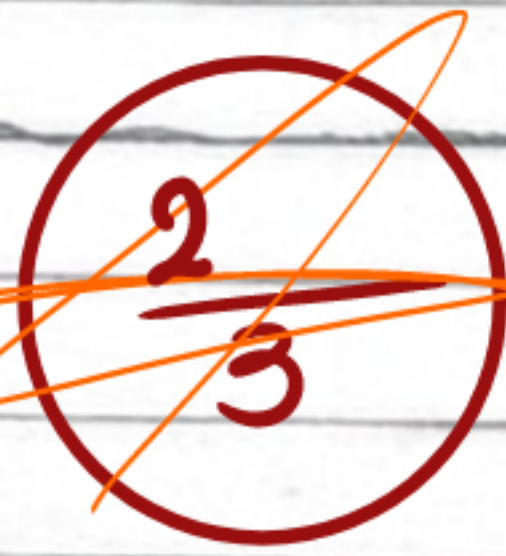
$$\therefore \text{Translation} = (6.5 - 4.5, 3.4 - 4.5) = (2, -1.1)$$

$$\therefore x' = 1.5x + 2, \quad y' = 1.5y - 1.1$$

$$\therefore \text{Matrix} = \begin{bmatrix} a & b & t_x \\ 0 & b & t_y \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1.5 & 0 & 2 \\ 0 & 1.5 & -1.1 \\ 0 & 0 & 1 \end{bmatrix}$$

→ Error in Calculation
 $S_y \neq t_y \quad -0.5$

→ Wrong approach for scale $S_y -0.5$



Q(3)

① $\oplus B - T_C - H_4$ ✓

② $\oplus C - T_D - H_1$ ✓

③ $\oplus E - T_B - H_3$ ✓

④ $\otimes A - T_E - H_5$ → دوجہ

⑤ $\oplus D - T_A - H_2$ ✓

Graph F represent
the intersection
Point between Negative
& Positive graph
image.

Reasons 2 marks , $\frac{1}{2}$ for wrong Graph ↓

$$\frac{5.5}{8}$$

Q(4)

V_{R0}	n_r	$P(r_k)$	CDF	n_2	P_2	CDF
0	1	$\frac{1}{32}$	$\frac{1}{32}$	0	0	0
1	4	$\frac{4}{32}$	$\frac{5}{32}$	0	0	0
2	0	$\frac{0}{32}$	$\frac{13}{32}$	4	$\frac{4}{32}$	$\frac{4}{32}$
3	5	$\frac{5}{32}$	$\frac{18}{32}$	4	$\frac{4}{32}$	$\frac{8}{32}$
4	2	$\frac{2}{32}$	$\frac{20}{32}$	4	$\frac{4}{32}$	$\frac{12}{32}$
5	2	$\frac{2}{32}$	$\frac{22}{32}$	4	$\frac{4}{32}$	$\frac{16}{32}$
6	5	$\frac{5}{32}$	$\frac{27}{32}$	5	$\frac{5}{32}$	$\frac{21}{32}$
7	5	$\frac{5}{32}$	1	11	$\frac{11}{32}$	1

~~3.25~~
5



In your paper (Image 1 out after map)

Image 1

1	7	6	5	0	2	2	2	1
2	7	6	4	1	2	1	3	7
3	6	6	3	2	2	1	3	7
4	6	5	3	2	3	2	4	7

Row

Column

V_{R0}	S_R	V_E	Z
0	0	0	0
1	1	0	3
2	3	0	4
3	4	2	5
4	4	3	6
5	5	4	7
6	6	5	7
7	7	7	7

-1

Wrong mapping

Output Image

7	7	6	0	4	4	4	3
7	7	5	3	4	3	5	7
7	7	5	4	4	3	5	7
7	6	5	4	5	4	5	7

Image 1

1	2	5	6	6	5	2	
1	3	5	6	6	6	3	1
1	3	6	6	6	6	3	1
		5	6	6	5	4	

-0.75

you moved pixels wrong!